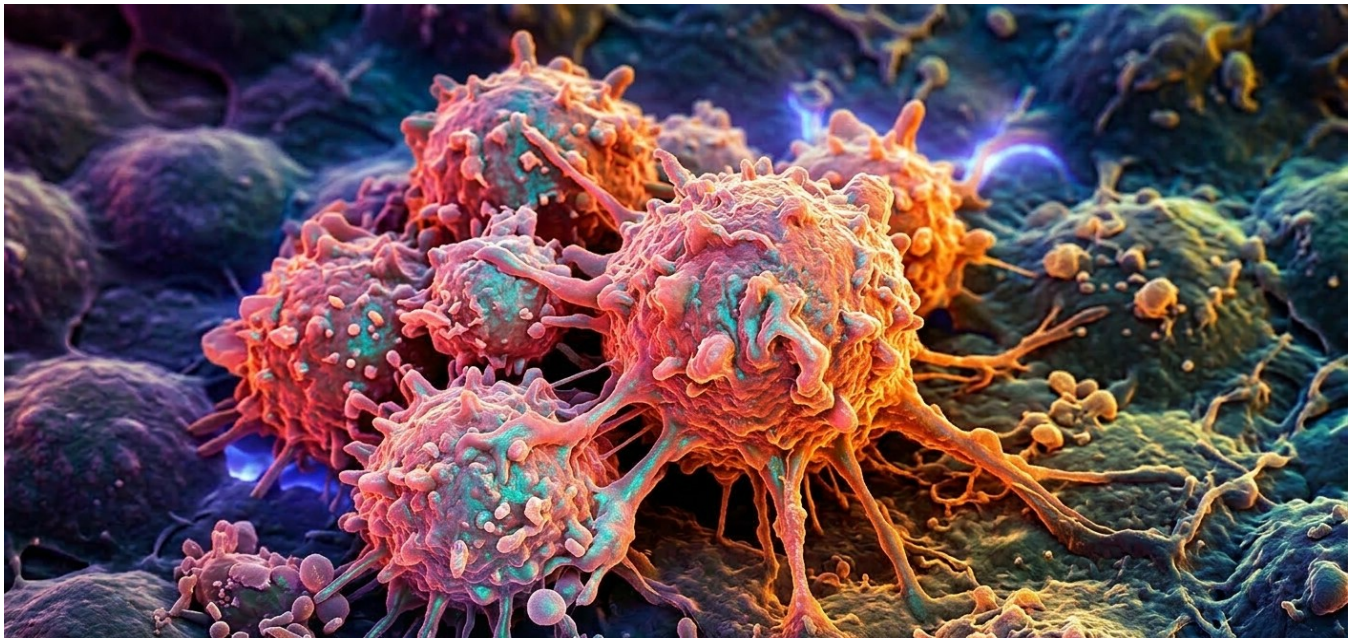

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Cancer Type Classification from RNA-Seq Gene Expression

A machine learning approach to identifying tumor tissue of origin from transcriptomic data

1. Overview & Motivation

When a cancer metastasizes across the body, one of the most critical and sometimes unanswerable clinical questions is: where did it originate? Identifying the tissue of origin directly determines treatment strategy. Breast cancer, kidney cancer, and lung cancer each require fundamentally different therapies; prescribing the wrong one not only fails to help the patient but can accelerate disease progression.

Historically, up to 3–5% of all cancer patients received a Cancer of Unknown Primary (CUP) diagnosis, meaning conventional histopathology and imaging could not identify the primary tumor site. Even with modern diagnostics, CUP still accounts for 1–2% of all cancers worldwide, representing tens of thousands of patients each year with limited treatment options.

To bridge this diagnostic gap, researchers are increasingly leveraging molecular profiling to identify the origin of these elusive cells through their internal signatures. RNA sequencing (RNA-Seq) offers a compelling solution. Every human cell maintains a distinct gene expression profile, a molecular fingerprint determined by which of its ~20,000 genes are active and at what level. Remarkably, even

after a cancer cell leaves its organ of origin and colonizes distant tissue, it retains much of this expression fingerprint. A metastatic breast cancer cell in the liver still behaves, transcriptionally, like a breast cell.

This project asks a direct question: can a machine learning model learn to classify 5 cancer types from raw RNA-Seq gene expression profiles alone, and can we identify which specific genes drive each classification decision?

2. Dataset

The dataset is sourced from The Cancer Genome Atlas (TCGA), accessed via Kaggle. It contains RNA-Seq expression profiles for 801 patients across 5 cancer types, with expression levels recorded for 20,531 genes per patient.

Code	Cancer Type	Tissue of Origin
BRCA	Breast Invasive Carcinoma	Breast (n=300)
KIRC	Kidney Renal Clear Cell Carcinoma	Kidney (n=146)
COAD	Colon Adenocarcinoma	Colon (n=78)
LUAD	Lung Adenocarcinoma	Lung (n=141)
PRAD	Prostate Adenocarcinoma	Prostate (n=136)

3. Methodology

3.1 Pre-processing

Variance filtering was applied first, reducing the feature space from 20,531 to 19,536 informative genes. Data was split 80/20 (stratified), StandardScaler was fit on the training set only, and SMOTE was applied to equalize all 5 cancer classes in the training distribution. Models were trained in both standard and SMOTE-balanced modes to measure the direct impact of balancing.

3.2 Dimensionality Reduction

PCA compressed 19,536 genes into 100 principal components (~75% variance) to reduce training cost and mitigate the curse of dimensionality. UMAP was additionally computed for visualization: its non-linear projection revealed 5 perfectly separated cancer clusters that PCA's linear projection could not resolve in 2D, confirming non-linear gene co-expression structure in the data.

3.3 Models & Evaluation

Four models were compared: Random Forest, SVM (RBF kernel), XGBoost, and MLP Neural Network. Each was evaluated on accuracy, ROC-AUC, and per-class F1. A 5-fold CV was performed within the training set to assess stability independently of the held-out test set.

4. Results

All four models achieved high classification accuracy. The table below summarizes performance across both training conditions on the held-out test set (160 samples).

Model	Accuracy (Standard)	Accuracy (+SMOTE)	ROC-AUC
Random Forest	96.89%	98.14%	0.9988
SVM (RBF)	98.14%	98.14%	0.9999
MLP Neural Net	98.76%	98.14%	0.9993
XGBoost	98.14%	96.27%	0.9979

Random Forest showed the greatest benefit from SMOTE (+1.24%), confirming that ensemble methods respond well to balanced distributions. SVM achieved identical accuracy in both conditions, indicating robustness to imbalance. XGBoost showed a slight decline with SMOTE (-1.86%), suggesting its built-in weighting mechanisms interact with oversampling in complex ways. All models achieved ROC-AUC above 0.997, with near-perfect class separation at the probability level.

5. Explainability AI/ SHAP Analysis

Accuracy alone does not answer the biological question. To identify which genes drive each classification, SHAP (SHapley Additive exPlanations) analysis was performed using a separate XGBoost model retrained on the full 19,536-gene space without PCA compression. This design choice is critical: SHAP on PCA components would map to mathematical abstractions, not biological genes. By bypassing PCA for explainability, every SHAP value corresponds directly to a named gene.

Five complementary SHAP visualizations were generated for each cancer type: a global gene importance bar chart, per-class beeswarm plots showing expression direction, a gene-by-cancer-type heatmap, individual patient waterfall explanations, and a gene direction analysis identifying which genes are over-expressed versus under-expressed per cancer type. These outputs transform the classifier from a black box into a biologically interpretable discovery tool.

6. Conclusions & future works

This project demonstrates that RNA-Seq gene expression profiles contain sufficient molecular information to classify 5 cancer types with near-perfect accuracy. The combination of SMOTE balancing, PCA compression, multi-model comparison, and gene-level SHAP explainability produces a pipeline that is both performant and biologically interpretable.

Planned extensions include mapping anonymous gene IDs to HGNC symbols via Ensembl Biomart for direct validation of biomarkers in the literature, pathway enrichment analysis using KEGG/GO annotations, and integration of additional TCGA omics layers (DNA methylation, miRNA) for multi-omics classification.

This is Project 01 of a 5-project Bioprocess & Bioengineering × Machine Learning portfolio being developed throughout 2026. Subsequent projects will address bioprocess optimization, biomaterial design, synthetic gene circuit simulation, and metabolic network analysis.

Dataset: TCGA RNA-Seq via Kaggle ·Tools: Python, scikit-learn, XGBoost, SHAP, UMAP, imbalanced-learn · Fully open source

Reference: Rassy, E., & Pavlidis, N. (2020). Progress in refining the clinical management of cancer of unknown primary in the molecular era. Nature reviews Clinical oncology, 17(9), 541-554.

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